

Fetching and Visualizing Official Statistics with R

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Interfaces to Official Statistics

- Packages or set of classes and methods to read data and metadata documents through exchange frameworks
 - Use [R](#) (or [Python](#)) packages to read data from APIs, databases, and web pages
 - * Individual packages:
 - [eurostat](#): Access data from Eurostat
 - [OECD](#): Access data from the OECD API
 - General-purpose packages:
 - * [rdbnomics](#): Unified access to many economic databases (e.g. ECB, **Eurostat**, IMF, World Bank)

- Interface standards:
 - [SDMX](#): Statistical Data and Metadata Exchange format
 - [pxweb](#): Access to data sources using the PX-Web API (e.g. Statistics Sweden, Statistics Estonia)

DBnomics

- [DBnomics](#) is a database of databases
 - free platform to aggregate publicly-available economic data provided by national and international statistical institutions, but also by researchers and private companies
 - Unified interface to access data from many sources
 - Harmonized data formats and metadata
 - Data series are available upon release by the provider
 - Each revision is archived to build a real-time database

How to fetch data (from DBnomics using R)

- DBnomics R client

```
install.packages("rdbnomics")  
library(rdbnomics)
```

Packages used in this tutorial

- Fetching data (`rdbnomics`)
- Data wrangling and transformation (`tidyverse`)
- Visualization (`ggplot2`, `plotly`)
- Tabular summaries (`gt`)
- Building this presentation (`quarto`)

```
library(quarto)      # for compiling Quarto presentations  
library(rdbnomics)   # for accessing economic data via DBnomics  
library(pxweb)  
library(tidyverse)   # dplyr, ggplot2, readr, etc.  
library(plotly)      # interactive visualizations  
library(gt)          # pretty tables  
library(countrycode)
```

Example: Fetch Unemployment Data

- Assume we know exactly the series ID we want to fetch
 - Unemployment rate, ILO definition, total, Estonia, from Eurostat

```
unemp <- rdb(ids = "Eurostat/ei_lmhr_m/M.PC_ACT.SA.LM-UN-T-TOT.EE") # fetch data
```

```
glimpse(unemp)
```

Rows: 296

Columns: 22

\$ `@frequency`	<chr> "monthly", "monthly", "monthly", "mo~
\$ dataset_code	<chr> "ei_lmhr_m", "ei_lmhr_m", "ei_lmhr_m~
\$ dataset_name	<chr> "Unemployment rate (%) - monthly dat~
\$ freq	<chr> "M", "M", "M", "M", "M", "M", "M", "~
\$ geo	<chr> "EE", "EE", "EE", "EE", "EE", "EE", ~
\$ `Geopolitical entity (reporting)`	<chr> "Estonia", "Estonia", "Estonia", "Es~
\$ indexed_at	<dtm> 2024-10-31 15:26:51, 2024-10-31 15:~
\$ indic	<chr> "LM-UN-T-TOT", "LM-UN-T-TOT", "LM-UN~
\$ Indicator	<chr> "Unemployment according to ILO defin~
\$ observations_attributes	<chr> "OBS_FLAG,", "OBS_FLAG,", "OBS_FLAG,~
\$ original_period	<chr> "2000-02", "2000-03", "2000-04", "20~
\$ original_value	<chr> "14.9", "14.2", "14.5", "13.9", "14"~
\$ period	<date> 2000-02-01, 2000-03-01, 2000-04-01,~
\$ provider_code	<chr> "Eurostat", "Eurostat", "Eurostat", ~
\$ s_adj	<chr> "SA", "SA", "SA", "SA", "SA", "SA", ~
\$ `Seasonal adjustment`	<chr> "Seasonally adjusted data, not calen~
\$ series_code	<chr> "M.PC_ACT.SA.LM-UN-T-TOT.EE", "M.PC~
\$ series_name	<chr> "Monthly - Percentage of population ~
\$ `Time frequency`	<chr> "Monthly", "Monthly", "Monthly", "Mo~
\$ unit	<chr> "PC_ACT", "PC_ACT", "PC_ACT", "PC_AC~
\$ `Unit of measure`	<chr> "Percentage of population in the lab~
\$ value	<dbl> 14.9, 14.2, 14.5, 13.9, 14.0, 13.9, ~

```
colnames(unemp)
```

[1] "@frequency"	"dataset_code"
[3] "dataset_name"	"freq"
[5] "geo"	"Geopolitical entity (reporting)"
[7] "indexed_at"	"indic"
[9] "Indicator"	"observations_attributes"
[11] "original_period"	"original_value"
[13] "period"	"provider_code"
[15] "s_adj"	"Seasonal adjustment"
[17] "series_code"	"series_name"
[19] "Time frequency"	"unit"
[21] "Unit of measure"	"value"

```
# Extract source and series ID from the metadata
(source_name <- unique(unemp$dataset_code))
```

```
[1] "ei_lmhr_m"
```

```
(provider_code <- unique(unemp$provider_code))
```

```
[1] "Eurostat"
```

```
(country_name <- unique(unemp$`Geopolitical entity (reporting)`))
```

```
[1] "Estonia"
```

```
(series_id <- unique(unemp$series_code))
```

```
[1] "M.PC_ACT.SA.LM-UN-T-TOT.EE"
```

```
# Plot the data
p1 <- ggplot(unemp, aes(x = period, y = value)) +
  geom_line(color = "steelblue", linewidth = 1) +
  labs(
    title = paste("Unemployment Rate in ", country_name),
```

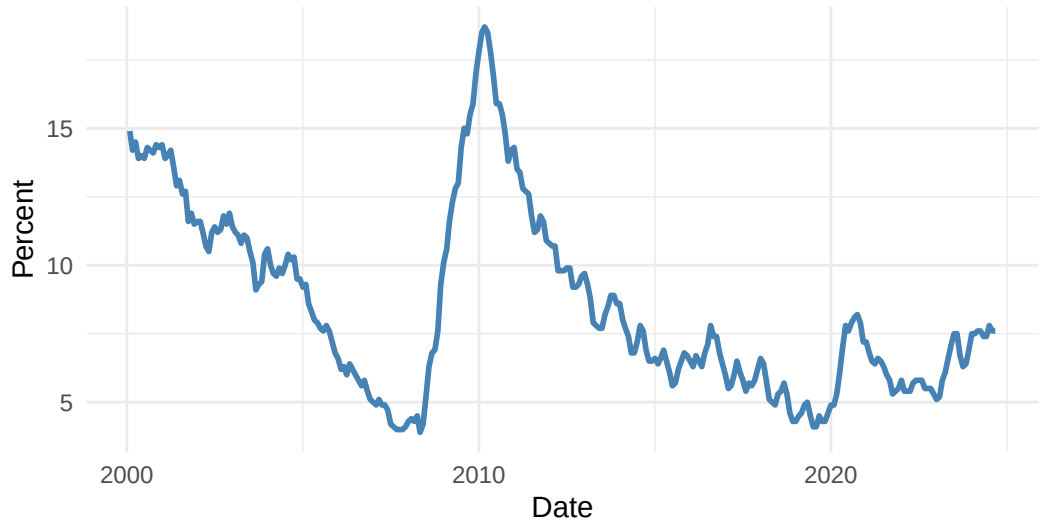
```

    subtitle = paste("Monthly, seasonally adjusted -", provider_code),
    x = "Date", y = "Percent",
    caption = paste("Source:", provider_code, "| Dataset:", source_name, "| ID:", series_id)
  ) +
  theme_minimal()
p1

```

Unemployment Rate in Estonia

Monthly, seasonally adjusted – Eurostat

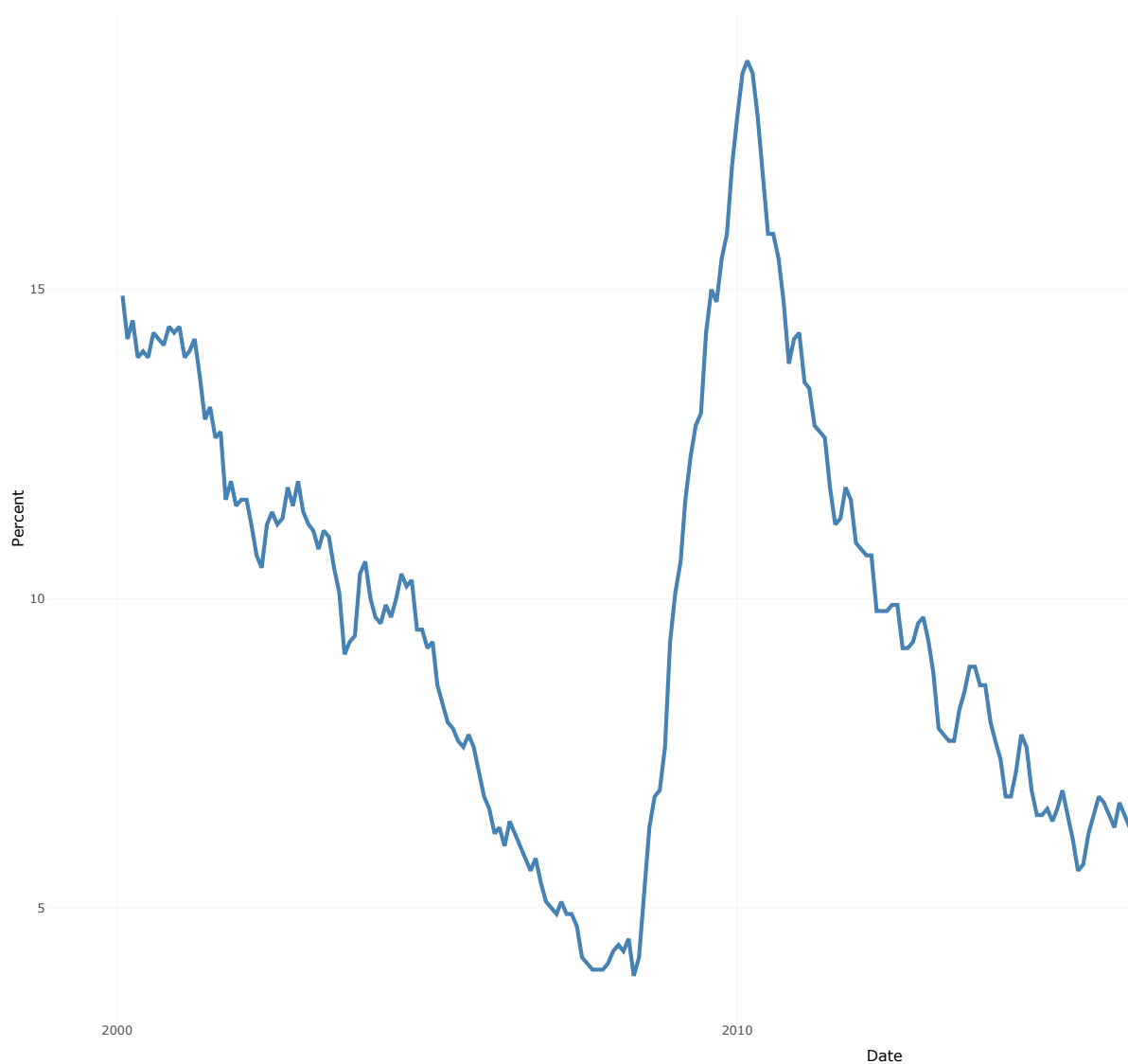


Source: Eurostat | Dataset: ei_lmhr_m | ID: M.PC_ACT.SA.LM-UN-T-TOT.EE

Interactive plot

```
ggplotly(p1)
```

Unemployment Rate in Estonia



```
source(here::here("scripts", "theme_ep.R"))
ggplot(unemp, aes(x = period, y = value)) +
  geom_line(color = "steelblue", linewidth = 1) +
  geom_hline(yintercept = 0, color = "#000", linewidth = 0.5) +
  scale_x_date(date_breaks = "1 year", date_labels = "%Y") +
  scale_y_continuous(labels = scales::label_number(suffix = "%")) +
  labs(
```

```

title = paste("Unemployment Rate in ", country_name),
subtitle = paste("Monthly, seasonally adjusted -", provider_code),
x = "",
x = "",
caption = paste(
  "Source:",
  provider_code,
  "| Dataset:",
  source_name,
  "| ID:",
  series_id
)
) +
theme_ep()

```



How do we find the series ID/mask/dimensions?

- Go to the [DBnomics website](#)
 - Search directly for a series or pick a provider

- Search for the [data](#) you want (dataset_code)
- Click on the [series](#) (series_code)
- Copy the series ID from the URL

-
- Show the available datasets of a provider:

```
head(rdb_datasets(provider_code = "Eurostat"))
```

```
$Eurostat
```

```

      code
      <char>
1:  aact_ali01
2:  aact_ali02
3:  aact_eaa01
4:  aact_eaa02
5:  aact_eaa03
---
8289: yth_empl_120
8290: yth_empl_130
8291: yth_empl_130
8292: yth_empl_140
8293: yth_empl_140

      name
      <char>
1:  Agricultural labour input statistics: absolute figures (1 000 annual work units)
2:  Agricultural labour input statistics: indices
3:  Economic accounts for agriculture - values at current prices
4:  Economic accounts for agriculture - values at n-1 prices
5:  Economic accounts for agriculture - values at constant prices (2015 = 100)
---
8289: Youth long-term unemployment rate (12 months or longer) by sex and age
8290: Youth long-term unemployment rate (12 months or longer) by sex, age and NUTS 2 regions
8291: Youth long-term unemployment rate (12 months or longer) by sex, age and NUTS 2 region
8292: Youth unemployment ratio by sex, age and NUTS 2 regions
8293: Youth unemployment ratio by sex, age and NUTS 2 region
```

-
- Show the dimensions of a dataset:


```
head(rdb_dimensions(provider_code = "Eurostat", dataset_code = "ei_lmhr_m"))
```

```
$Eurostat
$Eurostat$ei_lmhr_m
$Eurostat$ei_lmhr_m$freq
      freq Time frequency
      <char>      <char>
1:      M      Monthly
```

```
$Eurostat$ei_lmhr_m$geo
      geo      Geopolitical entity (reporting)
      <char>      <char>
1:      AT      Austria
2:      BA      Bosnia and Herzegovina
3:      BE      Belgium
4:      BG      Bulgaria
5:      CH      Switzerland
6:      CY      Cyprus
7:      CZ      Czechia
8:      DE      Germany
9:      DK      Denmark
10:     EA20     Euro area - 20 countries (from 2023)
11:      EE      Estonia
12:      EL      Greece
13:      ES      Spain
14: EU27_2020 European Union - 27 countries (from 2020)
15:      FI      Finland
16:      FR      France
17:      HR      Croatia
18:      HU      Hungary
19:      IE      Ireland
20:      IS      Iceland
21:      IT      Italy
22:      JP      Japan
23:      LT      Lithuania
24:      LU      Luxembourg
25:      LV      Latvia
26:      MT      Malta
27:      NL      Netherlands
28:      NO      Norway
29:      PL      Poland
30:      PT      Portugal
```

31:	RO	Romania
32:	SE	Sweden
33:	SI	Slovenia
34:	SK	Slovakia
35:	TR	Türkiye
36:	UK	United Kingdom
37:	US	United States
	geo	Geopolitical entity (reporting)

\$Eurostat\$ei_lmhr_m\$indic

indic

<char>

1:	LM-UN-F-GT25
2:	LM-UN-F-LE25
3:	LM-UN-F-TOT
4:	LM-UN-M-GT25
5:	LM-UN-M-LE25
6:	LM-UN-M-TOT
7:	LM-UN-T-GT25
8:	LM-UN-T-LE25
9:	LM-UN-T-TOT

Indicator

<char>

1:	Unemployment according to ILO definition - over 25 years - females
2:	Unemployment according to ILO definition - under 25 years - females
3:	Unemployment according to ILO definition - females
4:	Unemployment according to ILO definition - over 25 years - males
5:	Unemployment according to ILO definition - under 25 years - males
6:	Unemployment according to ILO definition - males
7:	Unemployment according to ILO definition - over 25 years - total
8:	Unemployment according to ILO definition - under 25 years - total
9:	Unemployment according to ILO definition - total

\$Eurostat\$ei_lmhr_m\$s_adj

s_adj

<char>

1:	NSA
2:	SA

Seasonal adjustment

<char>

1:	Unadjusted data (i.e. neither seasonally adjusted nor calendar adjusted data)
2:	Seasonally adjusted data, not calendar adjusted data

```

$Eurostat$ei_lmhr_m$unit
      unit
<char>
1: PC_ACT Percentage of population in the labour force

```

-
- Query to filter/select series from a provider's dataset

```

head(rdb_series(
  provider = "Eurostat",
  dataset_code = "ei_lmhr_m",
  query = "United Kingdom"
))

```

```

$Eurostat
$Eurostat$ei_lmhr_m
      series_code
      <char>
1: M.PC_ACT.NSA.LM-UN-F-GT25.UK
2: M.PC_ACT.NSA.LM-UN-F-LE25.UK
3:  M.PC_ACT.NSA.LM-UN-F-TOT.UK
4: M.PC_ACT.NSA.LM-UN-M-GT25.UK
5: M.PC_ACT.NSA.LM-UN-M-LE25.UK
6:  M.PC_ACT.NSA.LM-UN-M-TOT.UK
7: M.PC_ACT.NSA.LM-UN-T-GT25.UK
8: M.PC_ACT.NSA.LM-UN-T-LE25.UK
9:  M.PC_ACT.NSA.LM-UN-T-TOT.UK
10: M.PC_ACT.SA.LM-UN-F-GT25.UK
11: M.PC_ACT.SA.LM-UN-F-LE25.UK
12:  M.PC_ACT.SA.LM-UN-F-TOT.UK
13: M.PC_ACT.SA.LM-UN-M-GT25.UK
14: M.PC_ACT.SA.LM-UN-M-LE25.UK
15:  M.PC_ACT.SA.LM-UN-M-TOT.UK
16: M.PC_ACT.SA.LM-UN-T-GT25.UK
17: M.PC_ACT.SA.LM-UN-T-LE25.UK
18:  M.PC_ACT.SA.LM-UN-T-TOT.UK

```

1: Monthly - Percentage of population in the labour force - Unadjusted data (i.e. neither seasonally adjusted nor smoothed) - Unemployment according to ILO definition - over 25 years - females - United Kingdom

2: Monthly - Percentage of population in the labour force - Unadjusted data (i.e. neither seasonally adjusted nor calendar adjusted data) - Unemployment according to ILO definition - under 25 years - females - United Kingdom

3: Monthly - Percentage of population in the labour force - Unadjusted data (i.e. neither seasonally adjusted nor calendar adjusted data) - Unemployment according to ILO definition - females - United Kingdom

4: Monthly - Percentage of population in the labour force - Unadjusted data (i.e. neither seasonally adjusted nor calendar adjusted data) - Unemployment according to ILO definition - over 25 years - males - United Kingdom

5: Monthly - Percentage of population in the labour force - Unadjusted data (i.e. neither seasonally adjusted nor calendar adjusted data) - Unemployment according to ILO definition - under 25 years - males - United Kingdom

6: Monthly - Percentage of population in the labour force - Unadjusted data (i.e. neither seasonally adjusted nor calendar adjusted data) - Unemployment according to ILO definition - males - United Kingdom

7: Monthly - Percentage of population in the labour force - Unadjusted data (i.e. neither seasonally adjusted nor calendar adjusted data) - Unemployment according to ILO definition - over 25 years - total - United Kingdom

8: Monthly - Percentage of population in the labour force - Unadjusted data (i.e. neither seasonally adjusted nor calendar adjusted data) - Unemployment according to ILO definition - under 25 years - total - United Kingdom

9: Monthly - Percentage of population in the labour force - Unadjusted data (i.e. neither seasonally adjusted nor calendar adjusted data) - Unemployment according to ILO definition - total - United Kingdom

10: Monthly - Percentage of population in the labour force - Seasonally adjusted data, not calendar adjusted data - Unemployment according to ILO definition - United Kingdom

11: Monthly - Percentage of population in the labour force - Seasonally adjusted data, not calendar adjusted data - Unemployment according to ILO definition - United Kingdom

12: Monthly - Percentage of population in the labour force - Seasonally adjusted data, not calendar adjusted data - Unemployment according to ILO definition - United Kingdom

13: Monthly - Percentage of population in the labour force - Seasonally adjusted data, not calendar adjusted data - Unemployment according to ILO definition - United Kingdom

14: Monthly - Percentage of population in the labour force - Seasonally adjusted data, not calendar adjusted data - Unemployment according to ILO definition - United Kingdom

15: Monthly - Percentage of population in the labour force - Seasonally adjusted data, not calendar adjusted data - Unemployment according to ILO definition - United Kingdom

16: Monthly - Percentage of population in the labour force - Seasonally adjusted data, not calendar adjusted data - Unemployment according to ILO definition - United Kingdom

17: Monthly - Percentage of population in the labour force - Seasonally adjusted data, not calendar adjusted data - Unemployment according to ILO definition - United Kingdom

18: Monthly - Percentage of population in the labour force
Seasonally adjusted data, not calendar adjusted data - Unemployment according to ILO definition
United Kingdom

Fetch two (or more) series at once

- Example: Balance of Payments (BOP) for France and Germany from the [IMF](#) for Current Account, Total, Net, Euros, Millions, Annual

Option A

```
# by ID
bop <- rdb(ids = c("IMF/BOP/A.FR.BCA_BP6_EUR", "IMF/BOP/A.DE.BCA_BP6_EUR"))
bop %>% count(`Reference Area`)
```

	Reference Area	n
	<char>	<int>
1:	France	15
2:	Germany	26

Option B:

```
# by Mask
bop <- rdb(provider = "IMF",
           dataset_code = "BOP",
           mask = "A.FR+DE.BCA_BP6_EUR")
bop %>% count(`Reference Area`)
```

	Reference Area	n
	<char>	<int>
1:	France	15
2:	Germany	26

Option C:

You can not specify a dimension without a value!

```

# by Dimension
dim <- list(
  REF_AREA = c("DE", "FR"),
  INDICATOR = c("BCA_BP6_EUR"),
  FREQ = "A"
)
## Here I do not include FREQUENCY in the dimension list. I would download annual and quarterly
# dim <- list(
#   REF_AREA = c("DE", "FR"),
#   INDICATOR = c("BCA_BP6_EUR")
# )
bop <- rdb(provider = "IMF", dataset_code = "BOP", dimensions = dim)
bop %>% count(`Reference Area`)

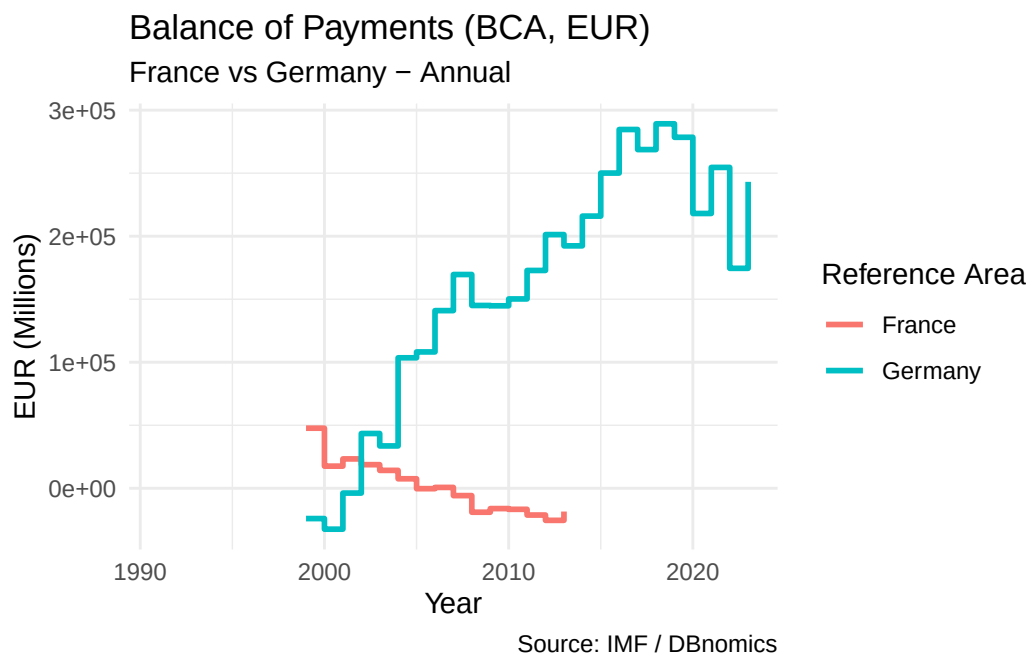
```

	Reference Area	n
	<char>	<int>
1:	France	15
2:	Germany	26

```

# Line plot with color by country
p2 <- ggplot(bop, aes(x = period, y = value, color = `Reference Area`)) +
  geom_step(linewidth = 1) +
  labs(
    title = "Balance of Payments (BCA, EUR)",
    subtitle = "France vs Germany - Annual",
    x = "Year",
    y = "EUR (Millions)",
    caption = "Source: IMF / DBnomics"
  ) +
  theme_minimal()
p2

```



Fetch two series from different datasets of different providers

```
unemp2 <- rdb(ids = c("AMECO/ZUTN/EA19.1.0.0.0.ZUTN", "Eurostat/une_rt_q/Q.SA.Y15-24.PC_ACT.7"))
```

```
# See which providers and datasets are included
dim(unemp2)
```

```
[1] 122 27
```

```
unique(unemp2$provider_code)
```

```
[1] "AMECO" "Eurostat"
```

```
unique(unemp2$dataset_code)
```

```
[1] "ZUTN" "une_rt_q"
```

```
unique(unemp2$series_code)
```

```
[1] "EA19.1.0.0.0.ZUTN"          "Q.SA.Y15-24.PC_ACT.T.EA19"
```

```
unique(unemp2$`@frequency`)
```

```
[1] "annual"      "quarterly"
```

```
unique(unemp2$`Seasonal adjustment`)
```

```
[1] NA  
[2] "Seasonally adjusted data, not calendar adjusted data"
```

```
# Summarize coverage and data availability  
unemp2_summary <- unemp2 %>%  
  group_by(series_code) %>%  
  summarize(  
    provider = first(provider_code),  
    dataset = first(dataset_code),  
    start_all = min(period, na.rm = TRUE),  
    end_all = max(period, na.rm = TRUE),  
    start_data = min(period[!is.na(value)]),  
    end_data = max(period[!is.na(value)]),  
    n_obs = sum(!is.na(value)),  
    .groups = "drop"  
  )
```

```
unemp2_summary_table <- unemp2_summary |>  
gt() %>%  
  tab_header(  
    title = "Time Coverage and Non-Missing Observations",  
    subtitle = "For Each Series from AMECO and Eurostat"  
  ) %>%  
  cols_label(  
    provider = "Provider",  
    dataset = "Dataset",  
    start_all = "Start (all)",  
    end_all = "End (all)",  
    start_data = "Start (data)",  
    end_data = "End (data)",  
    n_obs = "Number of observations",  
    .groups = "drop"
```


Time Coverage and Non-Missing Observations
For Each Series from AMECO and Eurostat

Series ID	Provider	Dataset	Start (all)	End (all)	Start (non-NA)	End (non-NA)
EA19.1.0.0.0.ZUTN	AMECO	ZUTN	1960-01-01	2026-01-01	1997-01-01	
Q.SA.Y15-24.PC_ACT.T.EA19	Eurostat	une_rt_q	2009-01-01	2022-07-01	2009-01-01	

```

series_code = "Series ID",
provider = "Provider",
dataset = "Dataset",
start_all = "Start (all)",
end_all = "End (all)",
start_data = "Start (non-NA)",
end_data = "End (non-NA)",
n_obs = "# Obs"
) %>%
fmt_date(
  columns = c(start_all, end_all, start_data, end_data),
  date_style = "iso"
) %>%
tab_options(
  table.width = pct(100),
  column_labels.font.weight = "bold"
)

```

```
unemp2_summary_table
```

```

# Metadata vectors
providers <- unique(unemp2$provider_code)
datasets <- unique(unemp2$dataset_code)
series_ids <- unique(unemp2$series_code)

```

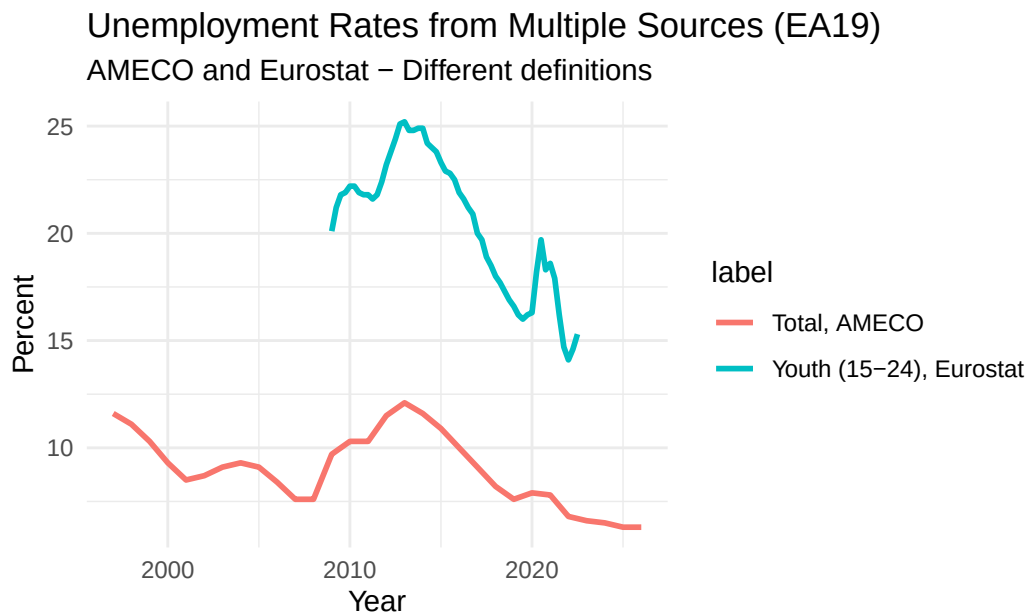
```

# Create a label that combines dataset + series ID
unemp2_clean <- unemp2 %>%

```

```
drop_na(value) %>%
mutate(label = case_when(
  series_code == "EA19.1.0.0.0.ZUTN" ~ "Total, AMECO",
  series_code == "Q.SA.Y15-24.PC_ACT.T.EA19" ~ "Youth (15-24), Eurostat",
  TRUE ~ series_code
))
```

```
p3 <- ggplot(unemp2_clean, aes(x = period, y = value, color = label)) +
  geom_line(linewidth = 1) +
  labs(
    title = "Unemployment Rates from Multiple Sources (EA19)",
    subtitle = "AMECO and Eurostat - Different definitions",
    x = "Year", y = "Percent",
    caption = paste("Series IDs:", paste(unique(unemp2_clean$series_code), collapse = " | "))
  ) +
  theme_minimal()
p3
```



Series IDs: EA19.1.0.0.0.ZUTN | Q.SA.Y15-24.PC_ACT.T.EA19

Fetch large amounts of data

```
cci_ee <- rdb("OECD", "DSD_HHDASH@DF_HHDASH_CTRY", "Q.EST.CCICP.IX")
unique(cci_ee$series_name)
```

```
[1] "Quarterly - Estonia - Consumer confidence - Index"
```

```
unique(cci_ee$REF_AREA)
```

```
[1] "EST"
```

Wildcarding dimensions

- To fetch **multiple values** for a dimension (e.g. countries), just **remove** the value from that position
 - Example: remove "EE" to fetch all countries (REF_AREA)

This can take a while

```
# cci_ee <- rdb("OECD", "DSD_HHDASH@DF_HHDASH_CTRY", "Q.EST.CCICP.IX")
cci <- rdb("OECD", "DSD_HHDASH@DF_HHDASH_CTRY", "Q..CCICP.IX")
# cci <- rdb("OECD", "DSD_HHDASH@DF_HHDASH_CTRY", "..CCICP.IX") # also wildcard the frequency
unique(cci$REF_AREA)
```

```
[1] "AUS"      "AUT"      "BEL"      "CHE"      "CHL"      "COL"
[7] "CRI"      "CZE"      "DEU"      "DNK"      "EA"       "ESP"
[13] "EST"      "EU27_2020" "FIN"      "FRA"      "G7"       "GBR"
[19] "GRC"      "HUN"      "IRL"      "ISR"      "ITA"      "JPN"
[25] "KOR"      "LTU"      "LUX"      "LVA"      "MEX"      "NLD"
[31] "NZL"      "OECD"     "POL"      "PRT"      "SVK"      "SVN"
[37] "SWE"      "TUR"      "USA"
```

```
cci_eu <- cci %>%
  filter(
    REF_AREA %in% c("EU27_2020", "AUT", "EST", "FIN", "DEU", "LVA", "LTU", "POL", "SWE"))
mutate(
  REF_AREA = case_when(
    REF_AREA == "EU27_2020" ~ "EU27",
```

```

      TRUE ~ countrycode(REF_AREA, origin = "iso3c", destination = "iso2c")
    )
  )
unique(cci_eu$`Reference area`)

```

```

[1] "Austria"
[2] "Germany"
[3] "Estonia"
[4] "European Union (27 countries from 01/02/2020)"
[5] "Finland"
[6] "Lithuania"
[7] "Latvia"
[8] "Poland"
[9] "Sweden"

```

```

library(gghighlight)
desired_order <- c(
  "European Union (27 countries from 01/02/2020)",
  "Austria",
  "Germany",
  "Estonia",
  "Finland",
  "Lithuania",
  "Latvia",
  "Poland",
  "Sweden"
)
unique(cci_eu$`Reference area`)

```

```

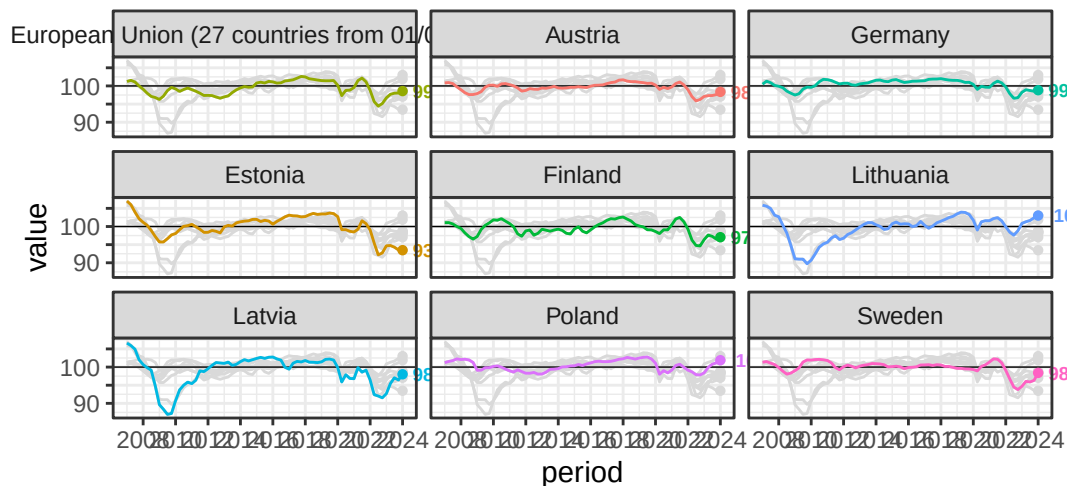
[1] "Austria"
[2] "Germany"
[3] "Estonia"
[4] "European Union (27 countries from 01/02/2020)"
[5] "Finland"
[6] "Lithuania"
[7] "Latvia"
[8] "Poland"
[9] "Sweden"

```

```

cci_eu |>
  ggplot() +
  geom_hline(yintercept = 100, linetype = "solid", size = .25) +
  geom_point(
    data = cci_eu %>%
      group_by(`Reference area`) %>%
      slice_max(period),
    aes(x = period, y = value, color = `Reference area`),
    shape = 16
  ) +
  geom_line(aes(x = period, y = value, color = `Reference area`)) +
  gghighlight(
    use_direct_label = FALSE,
    unhighlighted_params = list(colour = alpha("grey85", 1))
  ) +
  geom_text(
    data = cci_eu %>%
      group_by(`Reference area`) %>%
      slice_max(period),
    aes(x = period, y = value, color = `Reference area`, label = round(value)),
    hjust = -.5,
    vjust = .5,
    size = 2.5,
    fontface = "bold"
  ) +
  scale_x_date(
    date_labels = "%Y",
    date_breaks = "2 years"
  ) +
  scale_y_continuous(
    breaks = c(90, 95, 100, 105, 110),
    labels = c("90", "", "100", "", "110")
  ) +
  facet_wrap(
    ~ factor(`Reference area`, levels = desired_order)
  ) +
  coord_cartesian(clip = "off") +
  theme_bw() +
  theme(
    legend.position = "bottom"
  )

```



a — Austria a — European Union (27 countries from 01/02/2020) a — Germany a — Li
 area
 a — Estonia a — Finland a — Latvia a — Pr

- Sometimes you need to fetch many if not all dimensions of the data
- You can wildcard dimension and post-filter
- Example: MFI Interest Rate Statistics from the ECB
 - Start with a [single series](#) (Estonia, mortgage rates)

```
mir_mortgage_ee <- rdb("ECB", "MIR", "M.EE.B.A2C.A.R.A.2250.EUR.N")
unique(mir_mortgage_ee$series_name)
```

```
[1] "Monthly - Estonia - Deposit-taking corporations except the central bank (S.122) -
Lending for house purchase excluding revolving loans and overdrafts, convenience and extended
Total - Annualised agreed rate (AAR) / Narrowly defined effective rate (NDER) -
Total - Households and non-profit institutions serving households (S.14 and S.15) -
Euro - New business"
```

Wildcarding dimensions

- To fetch **multiple values** for a dimension (e.g. countries), just **remove** the value from that position

- Example: remove "EE" to fetch all countries (REF_AREA)

This can take a while

```
# mir_mortgage_ee <- rdb("ECB", "MIR", "M.EE.B.A2C.A.R.A.2250.EUR.N")
mir <- rdb("ECB", "MIR", "M..B..A.R.A..EUR.N")
unique(mir$REF_AREA)
```

```
[1] "AT" "BE" "CY" "DE" "EE" "ES" "FI" "FR" "GR" "HR" "IE" "IT" "LT" "LU" "LV"
[16] "MT" "NL" "PT" "SI" "SK" "U2"
```

```
unique(mir$BS_ITEM)
```

```
[1] "A2A" "A2AC" "A2B" "A2BC" "A2C" "A2CC" "A2D" "A2Z" "A2Z1" "A2Z3"
[11] "L21" "L22" "L23" "L24"
```

```
unique(mir$`BS counterpart sector`)
```

```
[1] "Non-Financial corporations (S.11)"
[2] "Households and non-profit institutions serving households (S.14 and S.15)"
[3] "Households of which sole proprietors and unincorporated partnerships (SP/UP)"
[4] "Non-Financial corporations and Households (S.11 and S.14 and S.15)"
```

Filter and plot

- Filter Estonia, Latvia and Lithuania
- Keep only selected **BS items** (loan categories)

```
# Filter by BS_ITEM and countries
mir_filtered <- mir %>%
  filter(
    REF_AREA %in% c("EE", "LV", "LT"),
    BS_ITEM %in% c("A2I", "A2C", "A2B", "A2J", "A2A")
  )
```

Plot interest rates by country & type

```
country_list <- paste(sort(unique(mir_filtered$REF_AREA)), collapse = ", ")
item_list <- paste(unique(mir_filtered$BS_ITEM), collapse = ", ")

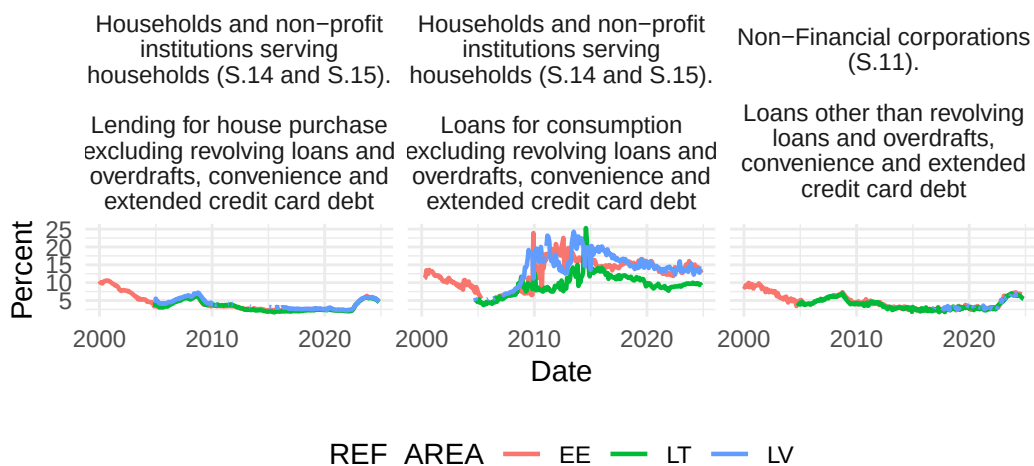
caption_text <- paste(
  "Source: ECB / DBnomics - Dataset code: MIR",
  paste0("\nFiltered: REF_AREA in ", country_list, "; BS_ITEM in ", item_list)
)

mir_filtered <- mir_filtered %>%
  mutate(facet_label = paste0(`BS counterpart sector`, ".\n\n", `Balance sheet item`))

p4 <- ggplot(mir_filtered, aes(x = period, y = value, color = REF_AREA)) +
  geom_line(linewidth = 0.8) +
  facet_wrap(~ facet_label, labeller = label_wrap_gen(width = 30), ncol = 3) +
  labs(
    title = "Interest Rates for Households and Firms",
    subtitle = "Faceted by Loan Type and Borrower Sector",
    x = "Date", y = "Percent",
    caption = caption_text
  ) +
  theme_minimal() +
  theme(legend.position = "bottom")
p4
```


Interest Rates for Households and Firms

Faceted by Loan Type and Borrower Sector



Source: ECB / DBnomics – Dataset code: MIR
Filtered: REF_AREA in EE, LT, LV; BS_ITEM in A2A, A2B, A2C

Accessing Statistics Estonia Data with pxweb

- [Statistics Estonia](#) is not yet part of DBnomics
- No dedicated R package for [andmed.stat.ee](#)
- But: you can access data using the `pxweb` R package
- Based on the **PX-WEB API**, which is also used by many Nordic stats agencies

How to Identify Variables

How to Identify Query Variables

- Use either:
 1. Programmatically via the metadata (`pxweb_get()`, `pxweb_get_metadata()`)
 2. From the Statistics Estonia web interface – click “**API query for this table**”

```
pa111_meta <- pxweb_get(  
  url = "https://andmed.stat.ee/api/v1/en/stat/PA111")  
pa111_meta
```

PXWEB METADATA

PA111: AVERAGE MONTHLY GROSS WAGES (SALARIES), MEDIAN, DECILES AND NUMBER OF EMPLOYEES BY ECONOMIC ACTIVITY
variables:

```
[[1]] Näitaja: Indicator  
[[2]] Tegevusala: Economic activity  
[[3]] Vaatlusperiood: Reference period
```

```
pa111_meta$variables[[1]]
```

\$code

```
[1] "Näitaja"
```

\$text

```
[1] "Indicator"
```

\$values

```
[1] "GR_W_AVG"      "NR_EMPL"      "GR_W_D1"      "GR_W_D2"      "GR_W_D3"  
[6] "GR_W_D4"      "GR_W_D5"      "GR_W_D6"      "GR_W_D7"      "GR_W_D8"  
[11] "GR_W_D9"      "GR_W_AVG_SM"
```

\$valueTexts

```
[1] "Average monthly gross wages (salaries), euros"  
[2] "Number of employees"  
[3] "1st decile of monthly gross wages (salaries), euros"  
[4] "2nd decile of monthly gross wages (salaries), euros"  
[5] "3rd decile of monthly gross wages (salaries), euros"  
[6] "4th decile of monthly gross wages (salaries), euros"  
[7] "Median (5th decile) of monthly gross wages (salaries), euros"  
[8] "6th decile of monthly gross wages (salaries), euros"  
[9] "7th decile of monthly gross wages (salaries), euros"  
[10] "8th decile of monthly gross wages (salaries), euros"  
[11] "9th decile of monthly gross wages (salaries), euros"  
[12] "Percentage change in average gross wages (salaries) compared to same period in previous year"
```

\$elimination

```
[1] FALSE
```

\$time

```
[1] FALSE
```

```

px_query_list1 <- list(
  "Näitaja" = c("GR_W_D1", "GR_W_D5", "GR_W_D9"),
  "Tegevusala" = "*",
  "Vaatlusperiood" = "*"
)

pa111 <- pxweb_get(
  url = "https://andmed.stat.ee/api/v1/en/stat/PA111",
  query = px_query_list1
)

# Convert to data frame
pa111.df <- as.data.frame(pa111, column.name.type = "text", variable.value.type = "text")
pa111.code <- as.data.frame(pa111, column.name.type = "code", variable.value.type = "code")
pa111.df$Näitaja <- pa111.code$Näitaja

```

```

selected_sectors <- c("Manufacturing", "Construction", "Information and communication", "Real estate and construction")
# Clean and reshape
pa111_plotdata <- pa111.df %>%
  mutate(
    quarter = yq(str_replace(`Reference period`, " ", "-")),
    Indicator = recode(
      Näitaja,
      "GR_W_D1" = "D1",
      "GR_W_D5" = "Median",
      "GR_W_D9" = "D9"
    )
  ) %>%
  filter(
    `Economic activity` %in% selected_sectors,
    Indicator %in% c("D1", "Median", "D9")
  ) %>%
  select(quarter, `Economic activity`, Indicator, value = `PA111: AVERAGE MONTHLY GROSS WAGES`)
  pivot_wider(names_from = Indicator, values_from = value)

```

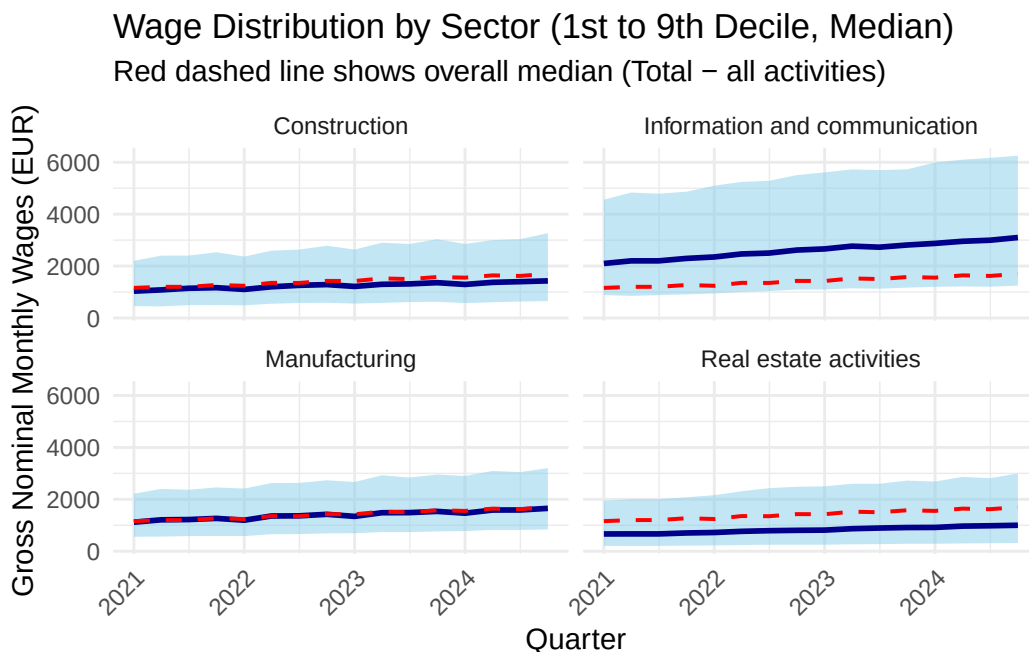
```

overall_median <- pa111_plotdata %>%
  filter(`Economic activity` == "Total - all activities") %>%
  select(quarter, total_median = Median)

```

```
pa111_plotdata <- pa111_plotdata %>%
  filter(`Economic activity` != "Total - all activities") %>%
  left_join(overall_median, by = "quarter")
```

```
ggplot(pa111_plotdata, aes(x = quarter)) +
  geom_ribbon(aes(ymin = D1, ymax = D9), fill = "skyblue", alpha = 0.5) +
  geom_line(aes(y = Median), color = "darkblue", linewidth = 1) +
  geom_line(aes(y = total_median), color = "red", linetype = "dashed", linewidth = 0.7) +
  facet_wrap(~`Economic activity`, ncol = 2) +
  labs(
    title = "Wage Distribution by Sector (1st to 9th Decile, Median)",
    subtitle = "Red dashed line shows overall median (Total - all activities)",
    x = "Quarter",
    y = "Gross Nominal Monthly Wages (EUR)"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
# library(DataExplorer)
# create_report(pa111_plotdata)
```